Uncertainty Quantification with Decision making for Real world application



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Introduction



Diabetes management requires continuous monitoring of blood glucose levels.



Continuous Glucose Monitoring (CGM) systems provide real-time, continuous tracking of blood glucose levels, enabling timely and informed diabetes management decisions.

ID 91507451
@ Piotr Adamowicz | Dreamstime.com

Issue Overview

Doctors receive continuous glucose data,
but lack reliable decision–support systems
to translate these values into optimal clinical actions.

Current approaches respond to glucose changes
after they occur instead of predicting and
preventing them.



Current systems either give raw values or predictions,
but without telling how much can trust them.

Objectives

To design a utility-based decision-making system that maps predicted glucose values to optimal clinical actions.

To move beyond simple prediction and build a system that can support real-world decisions.





Dataset

The dataset we have is a Time-series dataset where glucose values are recorded every 5 minutes.



We have 25 Patients and the data recorded is of minimum 14 days.
Unlike synthetic datasets, this dataset reflects real patient behavior, making the model suitable for practical deployment.



Woman Hand holding pencil and showing data on chart - By Karola G from Pexels {”https://www.canva.com/photos/MAEeTcufZfA/”}

Proposed Approach/Work Flow

Raw Time-series data

Deep-Learning model
LSTM/GRU

Predicting Future
Glucose Values

Prediction Intervals

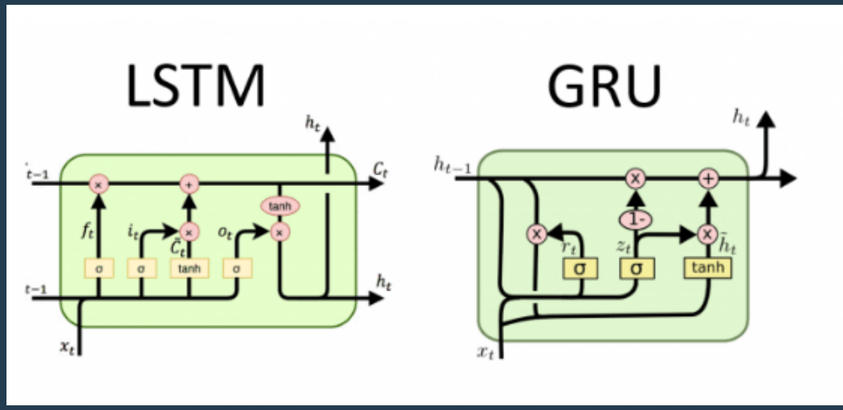
Utility Function Model
ANN + Sigmoid

Output Utility Score →
Action

Predicted Glucose +
Recommended Action



Evaluation Metrics



Model	PICP	MPIW
LSTM + Tube	0.9833	5.99
GRU + Tube	0.9802	4.30
TCN + Tube	0.9765	3.64

We aim for high PICP and low MPIW, and these results show a good balance between the two values.

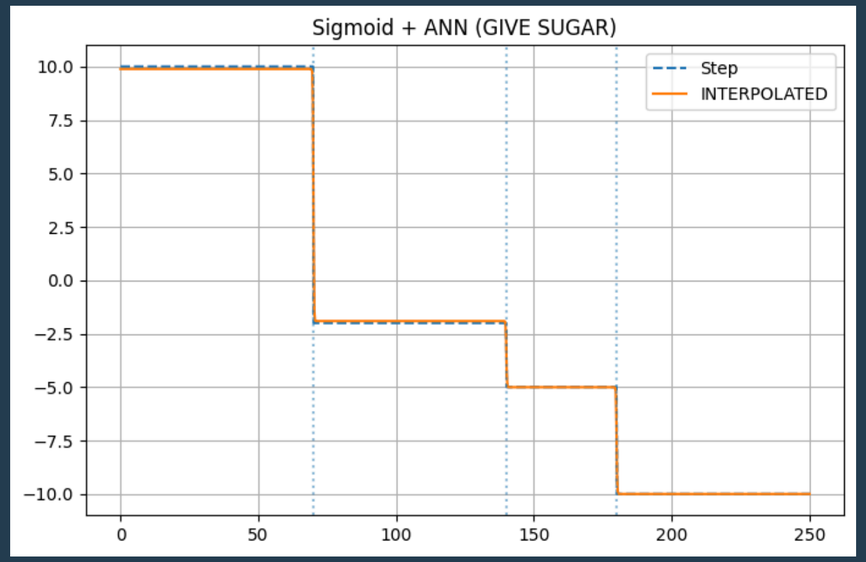
https://sh-tsang.medium.com/review-empirical-evaluation-of-gated-recurrent-neural-networks-on-sequence-modeling-gru-2adb86559257

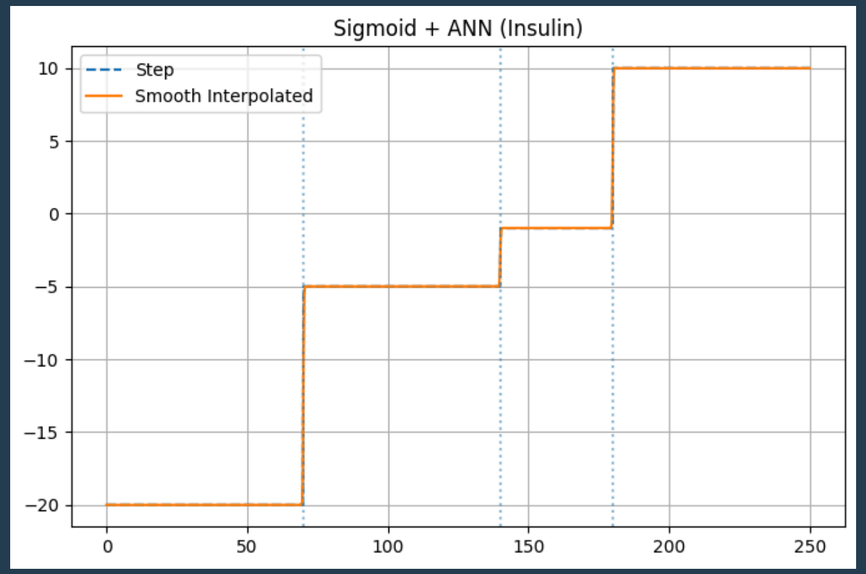
Methodology for generating Utility scores

Glucose Level	Action to be Taken	Example
Below 70 mg/DL	Take Sugar	Glucose Tablets, sugar
70-140	No Action	Nothing to be done
140-180	Monitor	Monitor consitently glucose levels
Above 180	Insulin	Give Insulin as per reqd. amount

Glucose Level	No Action	Sugar	Monitor	Insulin
Below 70	-10	+10	-2	-20
70-140	+5	-2	+2	-5
140-180	-2	-5	+5	-1
Above 180	-5	-10	-5	+10

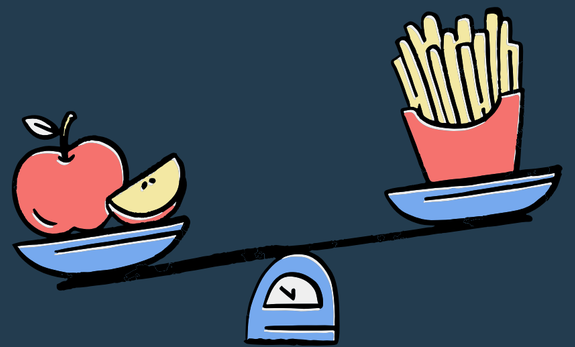
The static utility matrix is transformed into a self-learning Utility function using ANN and sigmoid modeling, enabling adaptive and data-driven decision-making.



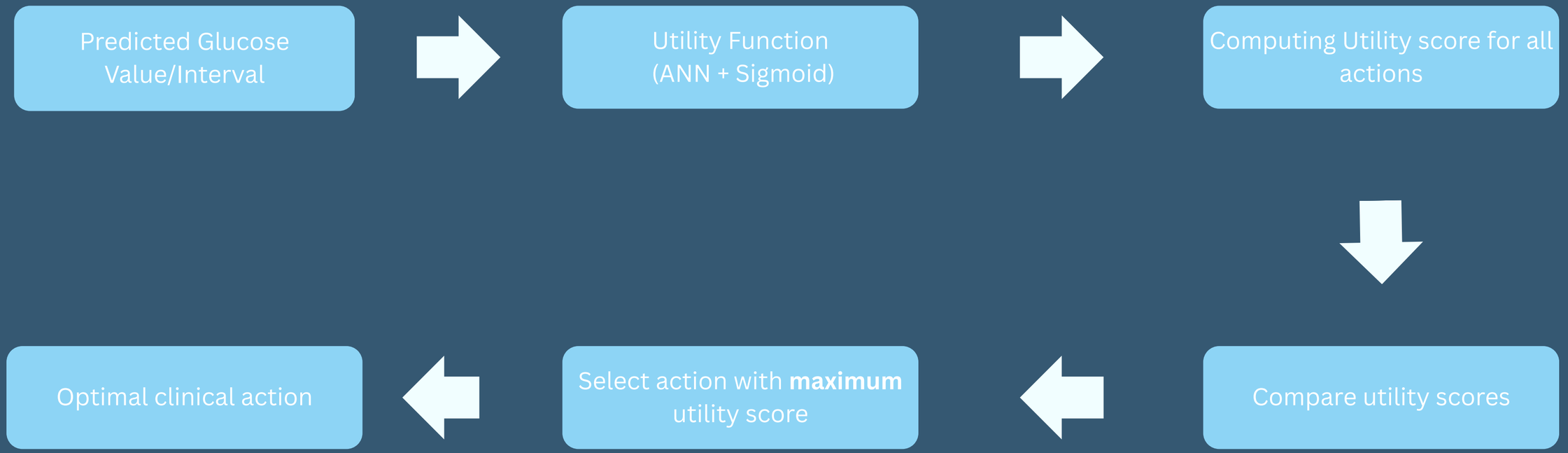


Utility Function

- The utility function is learned using an ANN, enabling data-driven mapping from glucose values to decision scores.
- Sigmoid activation ensures smooth, bounded outputs, avoiding abrupt transitions.
- This results in more realistic and stable decision-making compared to fixed threshold-based methods.



Algorithm for Selecing Best action



The ***optimal action*** is selected by computing utility scores for all the possible actions and choosing the one with the maximum utility.

Figure depicting Work Pipeline

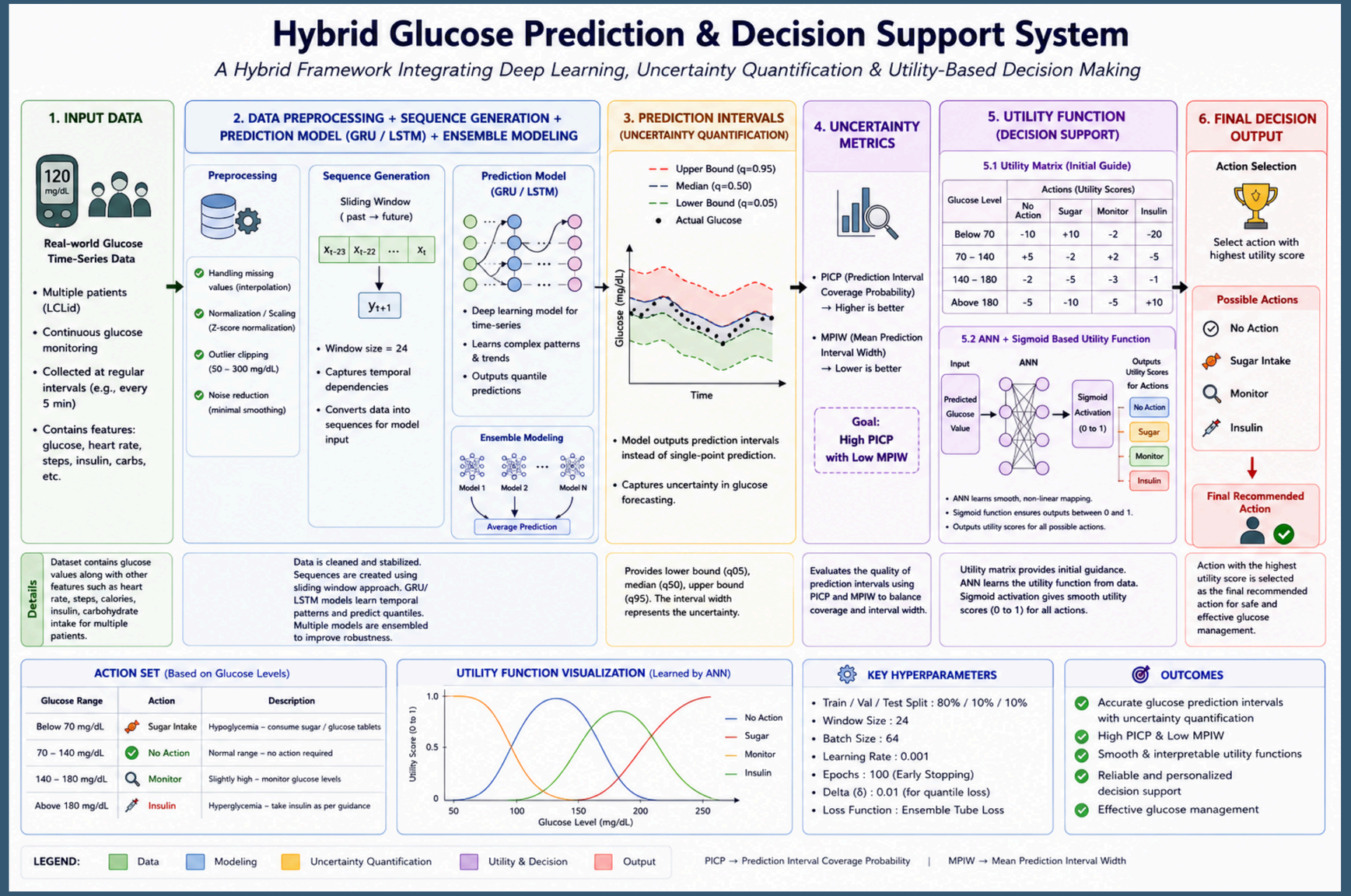
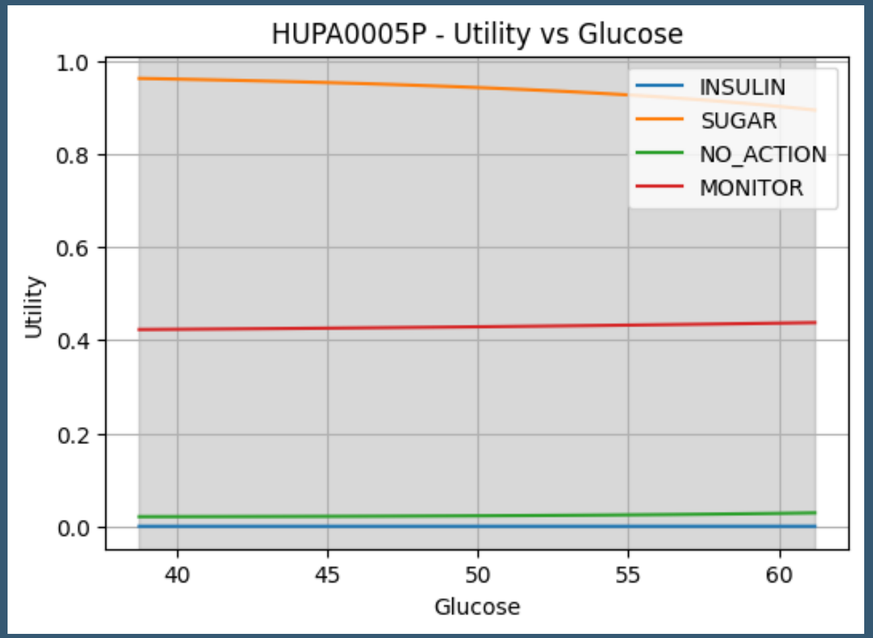
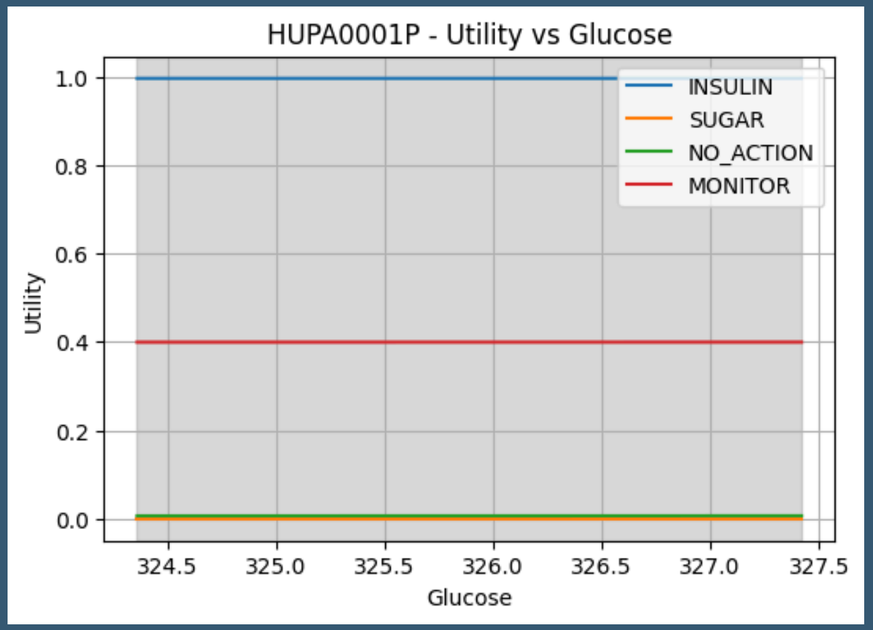


Image generated by ChatGPT AI

Results & Analysis

Patient_ID	Predicted Glucose Interval	U_Insulin	U_Sugar	U_No Action	U_Monitor	Best Action
HUPA0001P	342-327	10.5077	-9.4962	0.5328	0.4725	Insulin
HUPA0002P	116-125	-4.4996	-4.5240	2.4909	1.5155	No Action
HUPA0005P	38-63	-9.4869	10.4706	0.4898	0.5309	Sugar
HUPA0006P	157-183	-4.5020	-2.5191	2.4959	3.5005	Monitor





Demonstration of New Patient

New Patient Case:

A new patient arrives with CGM data recorded over the past few hours, along with recent insulin intake, carbohydrate consumption, and other activity levels.

System Processing:

The time-series data is preprocessed and passed through the trained LSTM model to predict future glucose levels.

The model outputs:

- Predicted Glucose: 162 mg/dL (True value we got by using CGM)
- Prediction Interval: [148-178] (Model output)

Demonstration of New Patient

Decision Layer:

The predicted values are fed into the ANN-based utility model, which computes the utility scores for all possible actions:

- **Insulin → 5.7236**
- **Monitor → 8.9219**
- **No Action → 3.4097**
- **Sugar → -2.1085**

Final Output:

The system recommends ***Monitor***, as it has the highest utility score, indicating the safest and most effective action under current conditions.

Learning Outcomes

- Implemented deep-learning models on Time-series data to capture temporal glucose dynamics.
- Transformed a static utility matrix into a self-learning utility function using ANN with sigmoid-based smoothing for realistic decision modeling.
- Developed an end-to-end decision pipeline that selects the optimal clinical action by maximizing utility across multiple actions.

CHALLENGES

- **Uncertainty in Glucose Dynamics:** Glucose levels are highly nonlinear and influenced by multiple factors like insulin, food intake, and activity, making accurate prediction difficult.

- **Balancing PICP and MPIW:** Achieving high reliability (coverage) while maintaining narrow prediction intervals is a key modeling challenge.

- **Lack of Ground Truth for Actions:** While glucose data is available, there is no direct mapping to optimal clinical actions, making decision modeling more complex.

CONCLUSION

- This work presents an integrated framework combining glucose prediction, uncertainty quantification, and utility-based decision-making.

- By using prediction intervals and ANN + sigmoid-based utility modeling, the system enables reliable and risk-aware clinical decisions.

- Overall, the approach moves beyond simple prediction towards intelligent, real-world decision-support in healthcare.

References

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Thank -You

